airtel

SYNC

Stadium Surge

Fix the anchor, then let the crowd shield the cells:
a ₹3.52 Cr, data-calibrated plan for a 1,00,000-fan final

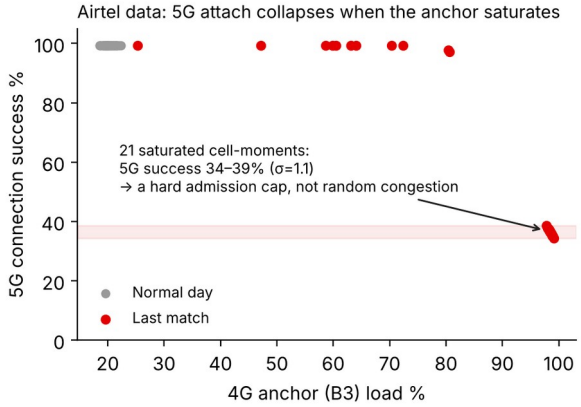
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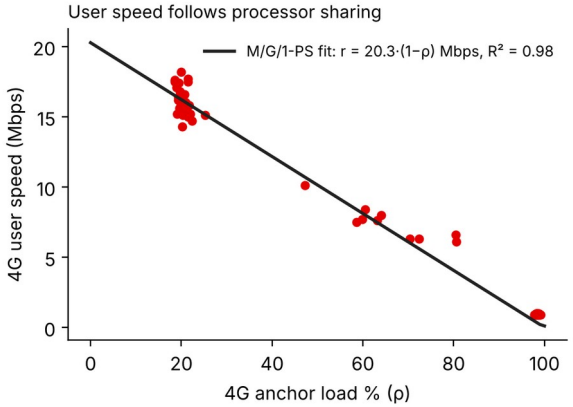


Platform Partner unstop

01 Diagnosis from Airtel's own counters: a hard cap on the 4G anchor

Sync_V1.xlsx, 8 stadium sites × 4 match moments, normal day vs last match (65k fans)





34–39%	5G attach success in all 21 saturated cell-moments (σ = 1.1). Flat = a configured cap, not random congestion
2.7×	attach demand vs anchor capacity at the innings break (from success = C/D)
R² = 0.98	Airtel 4G speeds follow M/G/1 processor sharing: r = 20.3·(1−ρ) Mbps
~24%	5G success at the final if nothing changes (demand × 1.54)

Key insight: 5G capacity was not the bottleneck. NSA 5G needs the B3 anchor to attach and carry its uplink.

- Proof: M6 (anchor 60%) kept 99.2% 5G success. Every cell above 97.8% anchor load sat at about 36%, a flat ceiling that points to an RRC-connected-user / admission limit on B3.
- Stuck 5G phones then add about 2,100 users to the full 4G layer. With processor sharing, speed → 20.3 × (1 − 0.985) ≈ 0.3–1 Mbps, exactly what was measured.
- So the plan fixes the anchor first (signalling) and adds capacity second (spectrum reuse).

02 Tasks 1 & 2: 31.7 Gbps peak → +15 5G / +27 4G cells → ₹3.52 Cr mix

Every cell planned at 80% load; traffic pushed to the most spectrally efficient layer (5G massive MIMO)

1,00,000	40,000	28k | 12k	5,040 | 2,160	25.2 | 6.5 Gbps
fans	Airtel (40%)	5G | 4G-only phones	online at peak (18%)	demand @ 5 | 3 Mbps

Cell type (DL)	Spectrum × SE	@80%	Users
5G macro 64T64R mMIMO	100 MHz × 74% × 16	947 Mbps	189 @5
5G small cell 4T4R	100 MHz × 74% × 7	414 Mbps	83 @5
4G macro sector	70 MHz × 2.5	140 Mbps	46 @3
4G small cell	40 MHz × 3.0	96 Mbps	32 @3

Requirement @80%	5G	4G
Peak demand	25.2 Gbps	6.5 Gbps
Cells needed (macro-eq.) / existing	27 / 12	47 / 20
Extra cells needed	**+15**	**+27**

Item	Qty	Cost	Adds
Dual-mode under-seat small cells	24 × ₹8 L	₹1.92 Cr	9.9 Gbps 5G + 2.3 4G
Cells on Wheels (gates, M4/M6 side)	4 × ₹25 L	₹1.00 Cr	12 × 5G + 12 × 4G sectors
Fibre ring 2 × 10G	1.2 km	₹0.60 Cr	no backhaul bottleneck
Reserve: licences, RRC uplift, rehearsal	—	₹0.48 Cr	10–12% contingency
Total		**₹4.00 Cr**	

Capacity check @80% (Gbps)	5G	4G
Existing + 4 COWs + 24 small cells	32.7	6.8
Peak load at the final	**62%**	**76%**

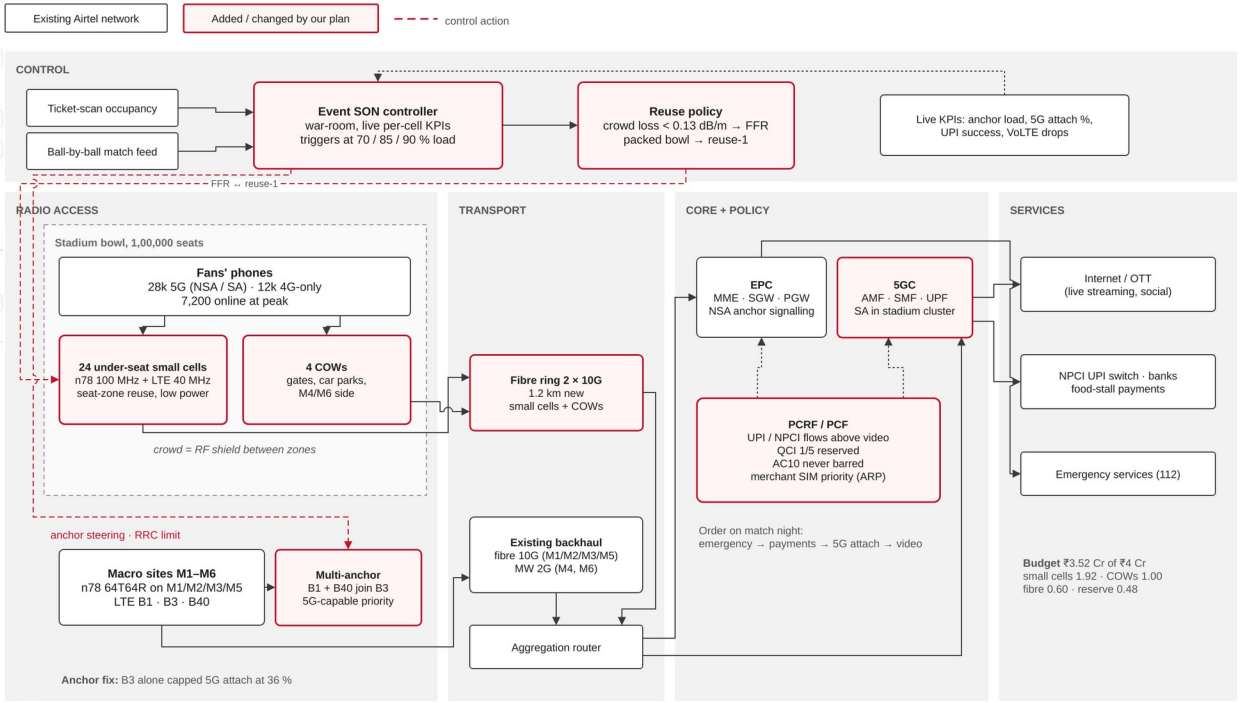
Why the anchor fix comes first

If 5G phones stay stuck (64% failure, as at the last match), 4G demand jumps from 6.5 to 16.2 Gbps: ~69 more 4G sectors ≈ 23 COWs ≈ ₹5.8 Cr, over budget. Keeping 5G phones on 5G costs ~₹0 in hardware, and 5G carries 6.8× more Mbps per cell. 73% of the spend (small cells + fibre) stays after the final.

03 Architecture: what changes in the network for match night

Thin outline = existing Airtel network · red outline = added by our plan · dashed red = live control action

Stadium Surge: network architecture for the final



Reading the diagram

- Control: an event SON controller reads ticket scans, the ball-by-ball feed and live KPIs
- Radio: 24 under-seat cells + 4 COWs; B1/B40 join B3 as NSA anchors
- Transport: new 2 × 10G fibre ring
- Core: SA in the stadium cluster; PCF puts UPI and emergency above video

Code + demo video

github.com/Amritha902/airtel-sync-stadium-surge

▶️ **Watch the 65-second demo**

MATLAB + Simulink + Python, editable draw.io source

04 Task 3: keep 5G phones on 5G, and protect payments and emergency calls

Root cause: the NSA B3 anchor ran out of RRC/PRB room, so 5G could not be added and 5G phones fell back to 4G

A. Unstick 5G (the anchor fix)

- Multi-anchor: add B1 and B40 as anchors so B3 is not a single choke point
- Reserve anchor room for 5G-capable phones; steer 4G-only phones to other bands (load balancing, SPID/RFSP)
- Lower the 5G add (B1) threshold and stop releasing 5G when the anchor is congested
- Raise the RRC-connected user licence and shorten inactivity timers during the match
- 5G SA on n78 in the stadium cluster for SA-capable phones, with no anchor needed

→ Target: 5G attach success of 95% or more (was 36%)

B. Protect UPI payments

- UPI uses only kilobytes but cannot wait: give UPI/NPCI and bank endpoints a high-priority QoS rule (PCC + DPI), above video
- Merchant SIMs at stalls get a priority profile (ARP/SPID) and dedicated concourse small cells
- Fallbacks: UPI Lite for small offline payments, UPI 123PAY over voice, and a stall Wi-Fi backup

→ Target: UPI failure under 1.5% (was 7.1%)

C. Guarantee emergency calls

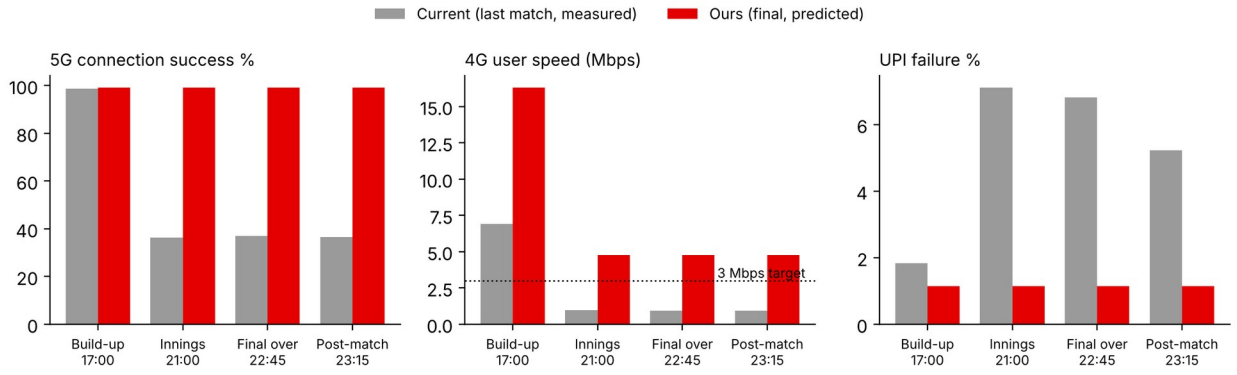
- Emergency (AC10) and VoLTE (QCI 1) are never barred; ARP pre-emption on
- Reserve about 5% of PRBs and RRC slots for QCI 1/5 at every stadium cell
- Priority SIMs (AC 11–15) for police, medical and the stadium ops team
- Access barring switched on automatically for normal users only when load passes 90%

→ Target: 100% emergency-call setup

Order of operations on match night: protect emergency calls first, then UPI payments, then 5G attach, then everything else. The things that matter most can never be pushed out by video.

05 Proof: current vs ours, on Airtel's own data

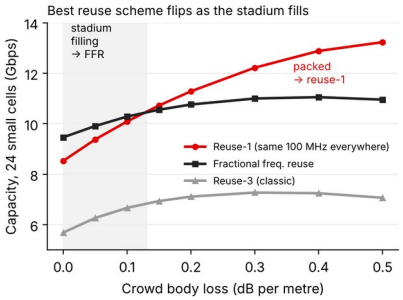
Left: last match (measured) vs final with our plan (predicted, same calibrated models). Bottom: our wireless simulations

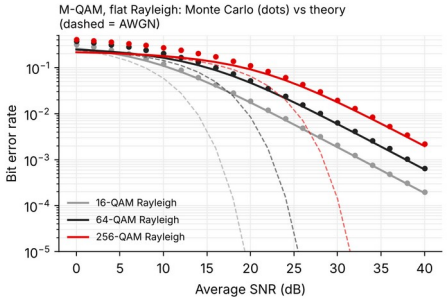


Novelty: the crowd is the shield

Human bodies attenuate 3.5 GHz (3GPP TR 38.901 §7.6.8 blockage). A packed bowl isolates under-seat cells, so the same 100 MHz can be reused in every seat zone.

The network switches FFR → reuse-1 as occupancy rises (ticket scans + ball-by-ball feed).





What the simulations show

- Packed stadium, 24 cells: reuse-1 gives 12.2 Gbps vs 11.0 (FFR) and 7.3 (reuse-3): +66% over classic planning
- Empty stadium: FFR wins, so the scheme must adapt
- Rayleigh M-QAM: Monte Carlo matches theory; 256-QAM needs ~25–30 dB, so SINR gain = more users on 256-QAM
- Python simulation (5 seeds, 5,040 users); MATLAB + Simulink versions included

06 Task 4: yes, there are side-effects. Here is how each one is contained

Most come from packing 40+ new cells into one bowl; all are controlled by design and a live war-room

Possible impact	Why it happens	Mitigation
Interference / lower SINR	Many cells in a small area	Low-power under-seat antennas, downtilt, PCI/PRB planning, tuned for MU-MIMO
Handover ping-pong, dropped calls	Fans move between macro, COW and small cell	Tune hysteresis and time-to-trigger; mobility-robustness SON; test at rehearsal match
Slower video for heavy users	Fair-share and priority rules at peak	Cap video at 720p only above 85% load; nothing below that; prioritise live score and payments
Brief access delay	Access barring for normal users	Only above 90% load, short barring timers, emergency never barred
Neighbourhood coverage changes	Re-tilted macros now face the stadium	Event-only tilt profile, switched on 3 h before and reverted after
Post-match crush at gates and metro	Traffic moves from bowl to exits	2 COWs at exits; capacity shifted by timetable
Extra phone battery drain	NSA dual connectivity (4G + 5G)	SA on n78 removes the dual link; DRX tuning

Live control: an event war-room

A per-cell dashboard (anchor load, 5G attach %, UPI success, VoLTE drops) with automatic triggers at 70/85/90% load. One rehearsal at a league match, 30 days before the final, plus one-click rollback for every parameter change. Triggers are timed to match phases: capacity is boosted 10 minutes before the innings break and the final over.

07 Competitive edge, impact and 60-day roadmap

How mega-events are usually handled, compared with this plan

Approach	Who typically uses it	Gap	This plan
Many COWs everywhere	Indian operators at IPL / big events	Costly every event; interference	4 COWs only, at exits and the weak side
Permanent DAS / in-venue small cells	US operators at Super Bowl venues	High capex, long lead time	24 under-seat cells now, with adaptive reuse
5G SA, no LTE anchor	Jio (SA network in India)	Needs SA core and SA phones	SA in the stadium + NSA multi-anchor
Purpose-built 5G venues	Qatar World Cup 2022 operators	Years of planning	60 days, ₹4 Cr, calibrated on real counters

36% → 99%
5G attach

1 → 4.8 Mbps
4G speed

12 → 70+ Mbps
5G speed

7.1% → ~1.2%
UPI failure

100% setup
Emergency

D-60 Plan & survey

D-45 Fibre + 24 cells

D-30 League-match rehearsal

D-7 COWs + tilt profile

D+1 Keep assets, playbook

Fix the 4G anchor first, then let the crowd shield the cells, so every Airtel fan is on the right layer and every payment and emergency call goes through.

Code, MATLAB/Simulink and demo video: github.com/Amritha902/airtel-sync-stadium-surge · ▶️ demo video